Atiyah–Macdonald, Chapter 5: Integral Dependence and Valuations

Solutions to the exercises

Conventions. A ring homomorphism $f : A \to B$ is *integral* if B is integral over $f(A)$; for a subring $A \subseteq B$ this is the usual notion. From the text I use: 5.1–5.3 (the elements of B integral over A form a subring; $A[x_1, \dots, x_n]$ is a finitely generated A-module if the x_i are integral), 5.4 (transitivity), 5.5 (the integral closure is integrally closed), 5.6 (integrality passes to quotients and rings of fractions), 5.7 and 5.8 (for $A \subseteq B$ integral domains, B is a field iff A is; hence $\mathfrak{q}$ maximal $\iff$ $\mathfrak{q}^c$ maximal), 5.9 (incomparability), 5.10 (lying over), 5.11 (going-up), 5.12, 5.13, 5.14, 5.15, 5.16 (going-down), 5.18 (valuation rings are local and integrally closed, overrings of valuation rings are valuation rings), 5.21 (a maximal pair (B, g) is a valuation ring), 5.22 (the integral closure is the intersection of the valuation rings containing A), and 5.23. Also: Ch. 1, Ex. 17, 18, 21; Ch. 3, Ex. 18, 21, 22, 23, 26; 1.11 (prime avoidance); Ch. 2, Ex. 20, 21. A "local subring" of a field is a subring that is a local ring.

This chapter has no numbered in-text exercises; there are 35 end-of-chapter exercises. I note where the printed statement is only exposition (Exercise 17 as printed is a trivial statement, and I prove the weak Nullstellensatz it is meant to illustrate) or needs a hypothesis (Exercise 4 needs $\operatorname{char} k \neq 2$).

End-of-chapter exercises

Exercise 1

> **Problem.** Let $f : A \to B$ be an integral homomorphism of rings. Show that $f^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is a *closed* mapping, i.e. that it maps closed sets to closed sets. (This is a geometrical equivalent of (5.10).)

Solution. Let $V(\mathfrak{b})$ be a closed subset of $\operatorname{Spec} B$. We show $f^*(V(\mathfrak{b})) = V(\mathfrak{b}^c)$, which is closed. The inclusion $\subseteq$ is clear: if $\mathfrak{q} \supseteq \mathfrak{b}$ then $f^*(\mathfrak{q}) = \mathfrak{q}^c \supseteq \mathfrak{b}^c$. Conversely let $\mathfrak{p} \supseteq \mathfrak{b}^c$ be a prime of A. The induced map $\bar{f} : A/\mathfrak{b}^c \to B/\mathfrak{b}$ is injective, and $B/\mathfrak{b}$ is integral over $\bar{f}(A/\mathfrak{b}^c)$ by 5.6(i). The prime $\mathfrak{p}/\mathfrak{b}^c$ of $A/\mathfrak{b}^c$ is therefore the contraction of a prime of $B/\mathfrak{b}$ (5.10), i.e. there is a prime $\mathfrak{q} \supseteq \mathfrak{b}$ of B with $\mathfrak{q}^c/\mathfrak{b}^c = \mathfrak{p}/\mathfrak{b}^c$, so $\mathfrak{q}^c = \mathfrak{p}$. Hence $\mathfrak{p} \in f^*(V(\mathfrak{b}))$. □

Exercise 2

Problem. Let A be a subring of a ring B such that B is integral over A, and let $f : A \to \Omega$ be a homomorphism of A into an algebraically closed field Ω. Show that f can be extended to a homomorphism of B into Ω. [Use (5.10).]

Solution. Let $\mathfrak{p} = \operatorname{Ker} f$, a prime ideal (the image is a subring of a field). By 5.10 there is a prime $\mathfrak{q}$ of B with $\mathfrak{q} \cap A = \mathfrak{p}$. Then $A/\mathfrak{p} \hookrightarrow B/\mathfrak{q}$ is an inclusion of integral domains with $B/\mathfrak{q}$ integral over $A/\mathfrak{p}$ (5.6). Let $K = \operatorname{Frac}(A/\mathfrak{p})$ and $L = \operatorname{Frac}(B/\mathfrak{q})$. The injection $A/\mathfrak{p} \to \Omega$ induced by f extends to $K \to \Omega$. Every element of $B/\mathfrak{q}$ is integral, hence algebraic, over K, so L is generated by algebraic elements and L/K is an algebraic extension. Since Ω is algebraically closed, the embedding $K \to \Omega$ extends to an embedding $L \to \Omega$ (Zorn's lemma, the standard extension theorem). The composite $B \to B/\mathfrak{q} \to L \to \Omega$ restricts on A to $A \to A/\mathfrak{p} \to K \to \Omega$, which is f. □

Exercise 3

Problem. Let $f : B \to B'$ be a homomorphism of A-algebras, and let C be an A-algebra. If f is integral, prove that $f \otimes 1 : B \otimes_A C \to B' \otimes_A C$ is integral. (This includes (5.6) ii) as a special case.)

Solution. Let $g = f \otimes 1$. The ring $B' \otimes_A C$ is generated as an abelian group by elements $b' \otimes c = (b' \otimes 1)(1 \otimes c)$. Now $1 \otimes c = g(1 \otimes c)$ lies in $g(B \otimes C)$. If $b' \in B'$ satisfies $b'^n + f(b_1)b'^{n-1} + \cdots + f(b_n) = 0$ with $b_i \in B$, then

$$(b' \otimes 1)^n + g(b_1 \otimes 1)(b' \otimes 1)^{n-1} + \cdots + g(b_n \otimes 1) = 0,$$

so $b' \otimes 1$ is integral over $g(B \otimes C)$. By 5.3 the elements of $B' \otimes C$ integral over $g(B \otimes C)$ form a subring; it contains all $b' \otimes 1$ and all $1 \otimes c$, hence all products and sums of these, i.e. everything. □

Exercise 4

Problem. Let A be a subring of a ring B such that B is integral over A. Let $\mathfrak{n}$ be a maximal ideal of B and let $\mathfrak{m} = \mathfrak{n} \cap A$ be the corresponding maximal ideal of A. Is $B_\mathfrak{n}$ necessarily integral over $A_\mathfrak{m}$? [Consider the subring $k[x^2-1]$ of $k[x]$, where k is a field, and let $\mathfrak{n} = (x-1)$. Can the element $1/(x+1)$ be integral?]

Solution. *No.* Let k be a field of characteristic $\neq 2$ (in characteristic 2 one has $x+1 = x-1$, and the element below is not in $B_\mathfrak{n}$). Put $y = x^2 - 1$, $A = k[y] \subseteq B = k[x]$. Then y is transcendental, so A is a polynomial ring, and B is integral over A since $x^2 - y - 1 = 0$. Let $\mathfrak{n} = (x-1)$; then $\mathfrak{m} = \mathfrak{n} \cap A = (y)$ (a polynomial $g(y)$ vanishes at $x = 1$ iff $g(0) = 0$). $B_\mathfrak{n}$ is an $A_\mathfrak{m}$-algebra because $A - \mathfrak{m} \subseteq B - \mathfrak{n}$.

The element $z = 1/(x+1)$ lies in $B_\mathfrak{n}$ since $x + 1 \notin \mathfrak{n}$ (its value at $x = 1$ is $2 \neq 0$). We have $z = (1-x)/(1-x^2) = (x-1)/y$ in $k(x)$, and $z \notin k(y)$ because $[k(x) : k(y)] = 2$ and $x \notin k(y)$; the conjugate of z over $k(y)$ (apply $x \mapsto -x$) is $z' = -(x+1)/y$, so $z + z' = -2/y$, $zz' = -1/y$, and the minimal polynomial of z over $k(y)$ is

$$t^2 + \tfrac{2}{y}\, t - \tfrac{1}{y}.$$

$A_{\mathfrak{m}} = k[y]_{(y)}$ is a principal ideal domain, hence integrally closed in $k(y)$. If z were integral over $A_{\mathfrak{m}}$, then by 5.15 (with $\mathfrak{a} = (1)$, $B = k(x)$) the coefficients of its minimal polynomial would lie in $A_{\mathfrak{m}}$; but $1/y \notin A_{\mathfrak{m}}$. So z is not integral over $A_{\mathfrak{m}}$, and $B_{\mathfrak{n}}$ is not integral over $A_{\mathfrak{m}}$. □

Remark. The hypothesis $\operatorname{char} k \neq 2$ is really needed for this example: if $\operatorname{char} k = 2$ then, with $u = x - 1$ and $y = u^2$, every element $a(u)/b(u)$ of $B_{\mathfrak{n}}$ ($b(0) \neq 0$) equals $a(u)b(u)/b(u)^2$ with $b(u)^2 \in k[y] - (y)$, so it lies in $A_{\mathfrak{m}}[u]$, which is integral over $A_{\mathfrak{m}}$; thus there $B_{\mathfrak{n}}$ is integral over $A_{\mathfrak{m}}$.

Exercise 5

Problem. Let $A \subseteq B$ be rings, B integral over A.

(i) If $x \in A$ is a unit in B then it is a unit in A.

(ii) The Jacobson radical of A is the contraction of the Jacobson radical of B.

Solution. (i) $x^{-1} \in B$ is integral over A: $x^{-m} + a_1 x^{-m+1} + \cdots + a_m = 0$ with $a_i \in A$. Multiplying by x^{m-1} gives $x^{-1} = -(a_1 + a_2 x + \cdots + a_m x^{m-1}) \in A$.

(ii) Every maximal ideal $\mathfrak{m}$ of A is of the form $\mathfrak{n} \cap A$ with $\mathfrak{n}$ a maximal ideal of B: by 5.10 there is a prime $\mathfrak{n}$ with $\mathfrak{n} \cap A = \mathfrak{m}$, and it is maximal by 5.8. Conversely $\mathfrak{n} \cap A$ is maximal for every maximal $\mathfrak{n}$ of B (5.8). Hence

$$\mathfrak{R}(A) = \bigcap_{\mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{n}} (\mathfrak{n} \cap A) = \left(\bigcap_{\mathfrak{n}} \mathfrak{n} \right) \cap A = \mathfrak{R}(B) \cap A = \mathfrak{R}(B)^c. \qquad \square$$

Exercise 6

Problem. Let $B_1, \ldots, B_n$ be integral A-algebras. Show that $\prod_{i=1}^n B_i$ is an integral A-algebra.

Solution. Let $f_i : A \to B_i$ be the structure maps and $f = (f_i) : A \to B = \prod B_i$. Let $b = (b_1, \ldots, b_n) \in B$. For each i choose a monic $P_i \in A[t]$ with $P_i^{f_i}(b_i) = 0$, where P^{f_i} denotes the polynomial obtained by applying f_i to the coefficients. Let $P = P_1 P_2 \cdots P_n \in A[t]$, monic. The i-th component of $P^f(b)$ is $P^{f_i}(b_i) = P_i^{f_i}(b_i) \cdot \prod_{j \neq i} P_j^{f_i}(b_i) = 0$. So $P^f(b) = 0$ and b is integral over $f(A)$. □

Exercise 7

Problem. Let A be a subring of a ring B, such that the set $B - A$ is closed under multiplication. Show that A is integrally closed in B.

Solution. Suppose $x \in B$ is integral over A but $x \notin A$. Choose a monic equation $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ ($a_i \in A$) of smallest degree $n \geq 1$. If $n = 1$ then $x = -a_1 \in A$, a contradiction,

so $n \geq 2$. Put $y = x^{n-1} + a_1 x^{n-2} + \cdots + a_{n-1} \in B$. Then $xy = -a_n \in A$. If $y \notin A$, then $x, y \in B - A$ and so $xy \in B - A$, contradicting $xy \in A$. Hence $y \in A$. But then

$$x^{n-1} + a_1 x^{n-2} + \cdots + a_{n-2} x + (a_{n-1} - y) = 0$$

is a monic equation of degree $n - 1 \geq 1$ for x over A, contradicting the minimality of n. □

Exercise 8

Problem.

(i) Let A be a subring of an integral domain B, and let C be the integral closure of A in B. Let f, g be monic polynomials in $B[x]$ such that $fg \in C[x]$. Then f, g are in $C[x]$. [Take a field containing B in which the polynomials f, g split into linear factors: say $f = \prod(x - \xi_i)$, $g = \prod(x - \eta_j)$. Each ξ_i and each η_j is a root of fg, hence is integral over C. Hence the coefficients of f and g are integral over C.]

(ii) Prove the same result without assuming that B (or A) is an integral domain.

Solution. I give one proof, valid for arbitrary rings; it contains (i) (for a domain B one may of course use a splitting field of fg over $\operatorname{Frac} B$ instead of the ring B'' below).

Step 1: a ring $B'' \supseteq B$ over which f and g split. If $h \in B[x]$ is monic of degree $d \geq 1$, then $B' = B[x]/(h)$ is a free B-module on $1, x, \ldots, x^{d-1}$, so $B \to B'$ is injective, and the class ξ of x is a root of h. Dividing by the monic polynomial $x - \xi$ in $B'[x]$ gives $h = (x - \xi)h_1$ with h_1 monic of degree $d - 1$. Repeating this with h_1 in place of h, and then with g, we obtain a ring $B'' \supseteq B$ and $f = \prod(x - \xi_i)$, $g = \prod(x - \eta_j)$ in $B''[x]$.

Step 2. $fg \in C[x]$ is monic and vanishes at each ξ_i and each η_j (a root of f or g is a root of fg). So these elements of B'' are integral over C. By 5.3 the elements of B'' integral over C form a subring C''. The coefficients of f and g are, up to sign, elementary symmetric functions in the ξ_i (resp. η_j), hence lie in C''. They also lie in B. So they are elements of B integral over C, hence integral over A (5.4, as $A \subseteq C \subseteq B''$), hence in C (definition of C). Therefore $f, g \in C[x]$. □

Exercise 9

Problem. Let A be a subring of a ring B and let C be the integral closure of A in B. Prove that $C[x]$ is the integral closure of $A[x]$ in $B[x]$. [If $f \in B[x]$ is integral over $A[x]$, then $f^m + g_1 f^{m-1} + \cdots + g_m = 0$ ($g_i \in A[x]$). Let r be an integer larger than m and the degrees of $g_1, \ldots, g_m$, and let $f_1 = f - x^r$, so that $(f_1 + x^r)^m + g_1(f + x^r)^{m-1} + \cdots + g_m = 0$, or say $f_1^m + h_1 f_1^{m-1} + \cdots + h_m = 0$, where $h_m = (x^r)^m + g_1(x^r)^{m-1} + \cdots + g_m \in A[x]$. Now apply Exercise 8 to the polynomials $-f_1$ and $f_1^{m-1} + h_1 f_1^{m-2} + \cdots + h_{m-1}$.]

Solution. *$C[x]$ is integral over $A[x]$.* A polynomial $\sum c_i x^i \in C[x]$ is a sum of terms $c_i x^i$ with c_i integral over $A \subseteq A[x]$ and $x \in A[x]$, so each term is integral over $A[x]$, and so is the sum (5.3).

Every $f \in B[x]$ integral over $A[x]$ lies in $C[x]$. Let $f^m + g_1 f^{m-1} + \cdots + g_m = 0$ with $g_i \in A[x]$. Choose r larger than m, than $\deg f$, and than all $\deg g_i$. (Only $r > \deg f$ and $r > \deg g_i$ are used below.) Put $f_1 = f - x^r$. Substituting $f = f_1 + x^r$ and expanding in powers of f_1 gives

$$f_1^m + h_1 f_1^{m-1} + \cdots + h_{m-1} f_1 + h_m = 0, \qquad h_i \in A[x],$$

where $h_m = x^{rm} + g_1 x^{r(m-1)} + \cdots + g_m$ (the value of the left side at $f_1 = 0$). Since $\deg g_i < r$ we have $\deg(g_i x^{r(m-i)}) < rm$ for $i \geq 1$, so h_m is monic of degree rm. Put $F = f_1^{m-1} + h_1 f_1^{m-2} + \cdots + h_{m-1} \in B[x]$. Then

$$(-f_1) \cdot F = h_m.$$

Since $r > \deg f$, the polynomial $-f_1 = x^r - f$ is monic of degree r. Dividing the monic polynomial h_m of degree rm by the monic $-f_1$: the product of $-f_1$ with a polynomial F has leading coefficient equal to that of F and degree $r + \deg F$ (as $-f_1$ is monic); comparing with h_m gives $\deg F = r(m-1)$ and F monic. Now $-f_1$ and F are monic in $B[x]$ and their product h_m lies in $A[x] \subseteq C[x]$. By Exercise 8, $-f_1 \in C[x]$. Hence $f = x^r + f_1 \in C[x]$. □

Exercise 10

Problem. A ring homomorphism $f : A \to B$ is said to have the *going-up property* (resp. the *going-down property*) if the conclusion of the going-up theorem (5.11) (resp. the going-down theorem (5.16)) holds for B and its subring $f(A)$.

Let $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the mapping associated with f.

(i) Consider the following three statements: (a) f^* is a closed mapping; (b) f has the going-up property; (c) let $\mathfrak{q}$ be any prime ideal of B and let $\mathfrak{p} = \mathfrak{q}^c$; then $f^* : \mathrm{Spec}(B/\mathfrak{q}) \to \mathrm{Spec}(A/\mathfrak{p})$ is surjective. Prove that (a)⇒(b)⇔(c). (See also Chapter 6, Exercise 11.)

(ii) Consider the following three statements: (a′) f^* is an open mapping; (b′) f has the going-down property; (c′) for any prime ideal $\mathfrak{q}$ of B, if $\mathfrak{p} = \mathfrak{q}^c$, then $f^* : \mathrm{Spec}(B_\mathfrak{q}) \to \mathrm{Spec}(A_\mathfrak{p})$ is surjective. Prove that (a′)⇒(b′)⇔(c′). (See also Chapter 7, Exercise 23.)

[To prove that (a′)⇒(c′), observe that $B_\mathfrak{q}$ is the direct limit of the rings B_t where $t \in B - \mathfrak{q}$; hence, by Chapter 3, Exercise 26, we have $f^*(\mathrm{Spec}(B_\mathfrak{q})) = \bigcap_t f^*(\mathrm{Spec}(B_t)) = \bigcap_t f^*(Y_t)$. Since Y_t is an open neighborhood of $\mathfrak{q}$ in Y, and since f^* is open, it follows that $f^*(Y_t)$ is an open neighborhood of $\mathfrak{p}$ in X and therefore contains $\mathrm{Spec}(A_\mathfrak{p})$.]

Solution. It suffices to treat chains of length 2: by induction on the lengths, the going-up (resp. going-down) property holds iff for primes $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ (resp. $\mathfrak{p}_1 \supseteq \mathfrak{p}_2$) of A and a prime $\mathfrak{q}_1$ of B with $\mathfrak{q}_1^c = \mathfrak{p}_1$ there is a prime $\mathfrak{q}_2 \supseteq \mathfrak{q}_1$ (resp. $\mathfrak{q}_2 \subseteq \mathfrak{q}_1$) with $\mathfrak{q}_2^c = \mathfrak{p}_2$. (Given a chain $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_n$ and $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$ one extends one step at a time.)

(i) (a)⇒(b). Let $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ and $\mathfrak{q}_1^c = \mathfrak{p}_1$. The set $f^*(V(\mathfrak{q}_1))$ is closed, and contains $\mathfrak{p}_1$; hence it contains the closure $V(\mathfrak{p}_1)$ of $\{\mathfrak{p}_1\}$ (Ch. 1, Ex. 18), in particular $\mathfrak{p}_2$. So $\mathfrak{p}_2 = \mathfrak{q}_2^c$ for some $\mathfrak{q}_2 \in V(\mathfrak{q}_1)$.

(b)⇔(c). The map f induces $\bar{f} : A/\mathfrak{p} \to B/\mathfrak{q}$ (for $\mathfrak{p} = \mathfrak{q}^c$), and under the identifications $\mathrm{Spec}(A/\mathfrak{p}) = V(\mathfrak{p})$, $\mathrm{Spec}(B/\mathfrak{q}) = V(\mathfrak{q})$ (Ch. 1, Ex. 21(iv)) the map $\bar{f}^*$ is $\mathfrak{q}' \mapsto \mathfrak{q}'^c$ (Ch. 3, Ex. 21(iii): $\bar{f}^*$ is the restriction of f^* to $V(\mathfrak{q})$). So (c) says: every prime $\mathfrak{p}' \supseteq \mathfrak{p}$ of A is the contraction of a prime $\mathfrak{q}' \supseteq \mathfrak{q}$ of B. This is exactly the going-up property for $\mathfrak{p} \subseteq \mathfrak{p}'$ and $\mathfrak{q}$ over $\mathfrak{p}$.

(ii) (b′)⇔(c′). Primes of $A_\mathfrak{p}$ correspond to primes $\mathfrak{p}' \subseteq \mathfrak{p}$ of A, primes of $B_\mathfrak{q}$ to primes $\mathfrak{q}' \subseteq \mathfrak{q}$ of B (3.13), and under these identifications $\mathrm{Spec}\, B_\mathfrak{q} \to \mathrm{Spec}\, A_\mathfrak{p}$ is $\mathfrak{q}' \mapsto \mathfrak{q}'^c$ (Ch. 3, Ex. 21(ii)). So (c′) says: every prime $\mathfrak{p}' \subseteq \mathfrak{p} = \mathfrak{q}^c$ is a contraction of a prime $\mathfrak{q}' \subseteq \mathfrak{q}$. This is going-down.

(a′)⇒(c′). Let $\mathfrak{q} \in Y = \operatorname{Spec} B$, $\mathfrak{p} = \mathfrak{q}^c$. The rings B_t, $t \in B - \mathfrak{q}$, with the restriction maps, form a direct system of A-algebras whose direct limit is $B_\mathfrak{q}$ (Ch. 3, Ex. 23(v)). By Ch. 3, Ex. 26,

$$f^*(\operatorname{Spec} B_\mathfrak{q}) = \bigcap_{t \notin \mathfrak{q}} f^*(\operatorname{Spec} B_t) = \bigcap_{t \notin \mathfrak{q}} f^*(Y_t),$$

since the image of $\operatorname{Spec} B_t$ in Y is the basic open set Y_t (Ch. 3, Ex. 21(i)). Each Y_t is an open neighbourhood of $\mathfrak{q}$, so as f^* is open, $f^*(Y_t)$ is an open neighbourhood of $\mathfrak{p}$ in $X = \operatorname{Spec} A$; by Ch. 3, Ex. 22 it contains the image of $\operatorname{Spec} A_\mathfrak{p}$. Hence the image of $\operatorname{Spec} B_\mathfrak{q}$ in $\operatorname{Spec} A$ contains the image of $\operatorname{Spec} A_\mathfrak{p}$, i.e. the set of primes $\mathfrak{p}' \subseteq \mathfrak{p}$. It is also contained there (if $\mathfrak{q}' \subseteq \mathfrak{q}$ then $\mathfrak{q}'^c \subseteq \mathfrak{p}$). So $\operatorname{Spec} B_\mathfrak{q} \to \operatorname{Spec} A_\mathfrak{p}$ is surjective. □

Exercise 11

> **Problem.** Let $f : A \to B$ be a flat homomorphism of rings. Then f has the going-down property. [Chapter 3, Exercise 18.]

Solution. Let $\mathfrak{q}$ be a prime of B and $\mathfrak{p} = \mathfrak{q}^c$. By Ch. 3, Ex. 18, $f^* : \operatorname{Spec}(B_\mathfrak{q}) \to \operatorname{Spec}(A_\mathfrak{p})$ is surjective. This is condition (c′) of Exercise 10, which is equivalent to the going-down property (b′). □

Exercise 12

> **Problem.** Let G be a finite group of automorphisms of a ring A, and let A^G denote the subring of G-invariants, that is of all $x \in A$ such that $\sigma(x) = x$ for all $\sigma \in G$. Prove that A is integral over A^G. [If $x \in A$, observe that x is a root of the polynomial $\prod_{\sigma \in G}(t - \sigma(x))$.]
>
> Let S be a multiplicatively closed subset of A such that $\sigma(S) \subseteq S$ for all $\sigma \in G$, and let $S^G = S \cap A^G$. Show that the action of G on A extends to an action on $S^{-1}A$, and that $(S^G)^{-1}A^G \cong (S^{-1}A)^G$.

Solution. *Integrality.* $P(t) = \prod_{\sigma \in G}(t - \sigma(x)) \in A[t]$ is monic and $P(x) = 0$ (the factor $\sigma = 1$ vanishes). For $\tau \in G$, τ applied to the coefficients permutes the factors ($\sigma \mapsto \tau\sigma$ is a bijection of G), so P has coefficients in A^G.

Extension of the action. Put $\sigma(a/s) = \sigma(a)/\sigma(s)$; this makes sense since $\sigma(s) \in S$. It is well defined: if $a/s = a'/s'$, i.e. $u(as' - a's) = 0$ for some $u \in S$, apply σ to get $\sigma(u)(\sigma(a)\sigma(s') - \sigma(a')\sigma(s)) = 0$ with $\sigma(u) \in S$. It is a ring homomorphism, and $\sigma \mapsto \sigma$ is an action ($\sigma\tau(a/s) = \sigma(\tau(a/s))$), so each σ is an automorphism.

The isomorphism. Define $\varphi : (S^G)^{-1}A^G \to S^{-1}A$ by $a/s \mapsto a/s$; its image lies in $(S^{-1}A)^G$ since $\sigma(a/s) = a/s$ for $a \in A^G$, $s \in S^G$. For $t \in S$ put $N(t) = \prod_{\sigma \in G} \sigma(t)$; then $N(t) \in S$ (since $\sigma(S) \subseteq S$ and S is multiplicatively closed) and $N(t) \in A^G$, so $N(t) \in S^G$, and $N(t) \in tA$.

Injective. If $a/s = 0$ in $S^{-1}A$ with $a \in A^G$, $s \in S^G$, then $ta = 0$ for some $t \in S$; hence $N(t)a = 0$ with $N(t) \in S^G$, so $a/s = 0$ in $(S^G)^{-1}A^G$.

Surjective onto $(S^{-1}A)^G$. Let $\xi = a/s \in (S^{-1}A)^G$. Put $s' = \prod_{\sigma \neq 1} \sigma(s) \in S$ and $t = ss' = N(s) \in S^G$; then $\xi = as'/t$. Since $\sigma(\xi) = \xi$ we get $\sigma(as')/t = as'/t$, so there is $u_\sigma \in S$

with $u_\sigma(\sigma(as') - as') = 0$. Let $u = \prod_\sigma u_\sigma \in S$ and $u' = N(u) \in S^G$; since $u \mid u'$ in A, $u'(\sigma(as') - as') = 0$ for all σ. Then $b = u'as' \in A$ satisfies $\sigma(b) = u'\sigma(as') = u'as' = b$, so $b \in A^G$ and $\xi = b/(u't)$ with $u't \in S^G$. □

Exercise 13

> **Problem.** In the situation of Exercise 12, let $\mathfrak{p}$ be a prime ideal of A^G, and let P be the set of prime ideals of A whose contraction is $\mathfrak{p}$. Show that G acts transitively on P. In particular, P is finite. [Let $\mathfrak{p}_1, \mathfrak{p}_2 \in P$ and let $x \in \mathfrak{p}_1$. Then $\prod_\sigma \sigma(x) \in \mathfrak{p}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{p}_2$, hence $\sigma(x) \in \mathfrak{p}_2$ for some $\sigma \in G$. Deduce that $\mathfrak{p}_1$ is contained in $\bigcup_{\sigma \in G} \sigma(\mathfrak{p}_2)$, and then apply (1.11) and (5.9).]

Solution. G acts on P: if $\mathfrak{P} \cap A^G = \mathfrak{p}$ then $\sigma(\mathfrak{P})$ is a prime of A and $\sigma(\mathfrak{P}) \cap A^G = \sigma(\mathfrak{P} \cap A^G) = \mathfrak{p}$ since σ fixes A^G pointwise.

Let $\mathfrak{p}_1, \mathfrak{p}_2 \in P$ and $x \in \mathfrak{p}_1$. The element $\prod_\sigma \sigma(x)$ lies in $\mathfrak{p}_1$ (the factor for $\sigma = 1$ is x), and is G-invariant; so it lies in $\mathfrak{p}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{p}_2$. As $\mathfrak{p}_2$ is prime, $\sigma(x) \in \mathfrak{p}_2$ for some σ, i.e. $x \in \sigma^{-1}(\mathfrak{p}_2)$. Hence $\mathfrak{p}_1 \subseteq \bigcup_{\sigma \in G} \sigma(\mathfrak{p}_2)$, a finite union of prime ideals. By 1.11, $\mathfrak{p}_1 \subseteq \sigma(\mathfrak{p}_2)$ for some σ. Both $\mathfrak{p}_1$ and $\sigma(\mathfrak{p}_2)$ contract to $\mathfrak{p}$ in A^G, and A is integral over A^G (Exercise 12); by 5.9 (incomparability), $\mathfrak{p}_1 = \sigma(\mathfrak{p}_2)$. So G is transitive on P, and P is a quotient of the finite set G, hence finite. □

Exercise 14

> **Problem.** Let A be an integrally closed domain, K its field of fractions and L a finite normal separable extension of K. Let G be the Galois group of L over K and let B be the integral closure of A in L. Show that $\sigma(B) = B$ for all $\sigma \in G$ and that $A = B^G$.

Solution. Let $b \in B$ satisfy a monic equation over A; applying $\sigma \in G$ (which fixes $K \supseteq A$) shows that $\sigma(b)$ satisfies the same monic equation, so $\sigma(b) \in B$. Thus $\sigma(B) \subseteq B$, and applying this to σ^{-1} gives $B \subseteq \sigma(B)$. So $\sigma(B) = B$.

Since L/K is Galois, $L^G = K$. Hence $B^G = B \cap L^G = B \cap K$. An element of $B \cap K$ is an element of K integral over A, so lies in A (A integrally closed). Conversely $A \subseteq B^G$. So $A = B^G$. □

Exercise 15

> **Problem.** Let A, K be as in Exercise 14, let L be any finite extension field of K, and let B be the integral closure of A in L. Show that, if $\mathfrak{p}$ is any prime ideal of A, then the set of prime ideals $\mathfrak{q}$ of B which contract to $\mathfrak{p}$ is finite (in other words, that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ has finite fibers). [Reduce to the two cases (a) L separable over K and (b) L purely inseparable over K. In case (a), embed L in a finite normal separable extension of K, and use Exercises 13 and 14. In case (b), if $\mathfrak{q}$ is a prime ideal of B such that $\mathfrak{q} \cap A = \mathfrak{p}$, show that $\mathfrak{q}$ is the set of all $x \in B$ such that $x^{p^m} \in \mathfrak{p}$ for some $m \geq 0$, where p is the characteristic of K, and hence that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is bijective in this case.]

Solution. Let L_s be the set of elements of L separable over K (a subfield). Then L/K factors as $K \subseteq L_s \subseteq L$ with L_s/K finite separable and L/L_s purely inseparable (if $\mathrm{char}\, K = 0$, $L_s = L$).

Let B_s be the integral closure of A in L_s; then $B_s = B \cap L_s$, $A \subseteq B_s \subseteq B$, and B_s is integrally closed in L_s (an element of L_s integral over B_s is integral over A by 5.4), with $\mathrm{Frac}(B_s) = L_s$ (any $v \in L_s$ satisfies $a_0 v^r + \cdots = 0$ with $a_0 \neq 0$, so $a_0 v \in B_s$). A prime $\mathfrak{q}$ of B lying over $\mathfrak{p}$ contracts to a prime $\mathfrak{q} \cap B_s$ of B_s lying over $\mathfrak{p}$. So it suffices to show (a): $\operatorname{Spec} B_s \to \operatorname{Spec} A$ has finite fibres, and (b): $\operatorname{Spec} B \to \operatorname{Spec} B_s$ has fibres with at most one point.

(a) Let $M \supseteq L_s$ be a finite normal separable extension of K (the normal closure of L_s; it is finite and separable), $G = \mathrm{Gal}(M/K)$ and C the integral closure of A in M. By Exercise 14, $\sigma(C) = C$ for $\sigma \in G$ and $C^G = A$, and $B_s = C \cap L_s$. Given a prime $\mathfrak{q}$ of B_s over $\mathfrak{p}$, by lying over (5.10) for $B_s \subseteq C$ there is a prime $\mathfrak{Q}$ of C with $\mathfrak{Q} \cap B_s = \mathfrak{q}$; then $\mathfrak{Q} \cap A = \mathfrak{p}$. By Exercise 13 (applied to G acting on C, with $C^G = A$) the set of primes of C over $\mathfrak{p}$ is finite. Since $\mathfrak{q}$ is obtained from such a $\mathfrak{Q}$ by $\mathfrak{Q} \mapsto \mathfrak{Q} \cap B_s$, the fibre of $\operatorname{Spec} B_s \to \operatorname{Spec} A$ over $\mathfrak{p}$ is finite.

(b) Here $\mathrm{char} = p > 0$, and every $x \in L$ has $x^{p^m} \in L_s$ for some $m \geq 0$. Let $\mathfrak{q}$ be a prime of B with $\mathfrak{q} \cap B_s = \mathfrak{p}_s$ (a prime of B_s). If $x \in \mathfrak{q}$, take m with $x^{p^m} \in L_s$; it is integral over B_s (5.3), hence in B_s, and in $\mathfrak{q}$, so $x^{p^m} \in \mathfrak{q} \cap B_s = \mathfrak{p}_s$. Conversely, if $x \in B$ and $x^{p^m} \in \mathfrak{p}_s \subseteq \mathfrak{q}$ then $x \in \mathfrak{q}$. So $\mathfrak{q} = \{x \in B : x^{p^m} \in \mathfrak{p}_s \text{ for some } m\}$ is determined by $\mathfrak{p}_s$; there is at most one such $\mathfrak{q}$ (and exactly one by 5.10).

Combining, the fibre of $\operatorname{Spec} B \to \operatorname{Spec} A$ over $\mathfrak{p}$ has at most as many points as the (finite) fibre of $\operatorname{Spec} B_s \to \operatorname{Spec} A$ over $\mathfrak{p}$. □

Exercise 16

> **Problem.** (*Noether's normalization lemma.*) Let k be a field and let $A \neq 0$ be a finitely generated k-algebra. Then there exist elements $y_1, \ldots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \ldots, y_r]$.
>
> [Printed text: We shall assume that k is *infinite*. (The result is still true if k is finite, but a different proof is needed.) Let $x_1, \ldots, x_n$ generate A as a k-algebra. We can renumber the x_i so that $x_1, \ldots, x_r$ are algebraically independent and each of $x_{r+1}, \ldots, x_n$ is algebraic over $k[x_1, \ldots, x_r]$. Now proceed by induction on n. If $n = r$ there is nothing to do, so suppose $n > r$ and the result true for $n-1$ generators. The generator x_n is algebraic over $k[x_1, \ldots, x_{n-1}]$, hence there exists a polynomial $f \neq 0$ in n variables such that $f(x_1, \ldots, x_{n-1}, x_n) = 0$. Let F be the homogeneous part of highest degree in f. Since k is infinite, there exist $\lambda_1, \ldots, \lambda_{n-1} \in k$ such that $F(\lambda_1, \ldots, \lambda_{n-1}, 1) \neq 0$. Put $x_i' = x_i - \lambda_i x_n$ $(1 \leq i \leq n-1)$. Show that x_n is integral over the ring $A' = k[x_1', \ldots, x_{n-1}']$, and hence that A is integral over A'. Then apply the inductive hypothesis to A'.
>
> From the proof it follows that $y_1, \ldots, y_r$ may be chosen to be linear combinations of $x_1, \ldots, x_n$. This has the following geometrical interpretation: if k is algebraically closed and X is an affine algebraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace L of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L. [Use Exercise 2.]]

Solution. The normalization lemma for arbitrary k. Let $A \neq 0$ be generated by $x_1, \ldots, x_n$. Induct on n. If $n = 0$, $A = k \cdot 1 \cong k$ (it is a non-zero quotient of the field k), and we take $r = 0$. Let $n \geq 1$. If $x_1, \ldots, x_n$ are algebraically independent, take $y_i = x_i$. Otherwise there is a non-zero polynomial $f \in k[t_1, \ldots, t_n]$ with $f(x_1, \ldots, x_n) = 0$. Let e be an integer larger than every exponent occurring in f. For a monomial $t^\alpha = t_1^{\alpha_1} \cdots t_n^{\alpha_n}$ put $N(\alpha) = \alpha_n + \alpha_1 e + \alpha_2 e^2 + \cdots + \alpha_{n-1} e^{n-1}$; by uniqueness of base-$e$ expansion, N is injective on the exponents

occurring in f. Substitute $t_i = t'_i + t_n^{e^i}$ $(1 \le i \le n-1)$: the monomial t^α becomes $t_n^{N(\alpha)}$ + (terms of lower degree in t_n with coefficients in $k[t']$). Hence

$$f\big(t'_1 + t_n^e, \ldots, t'_{n-1} + t_n^{e^{n-1}}, t_n\big) = c\, t_n^{N^*} + (\text{lower terms in } t_n \text{ with coefficients in } k[t'_1, \ldots, t'_{n-1}]),$$

where N^* is the largest value of $N(\alpha)$ and $c \in k$ is the (non-zero) coefficient of the corresponding monomial of f. Put $x'_i = x_i - x_n^{e^i}$ and $A' = k[x'_1, \ldots, x'_{n-1}]$. Evaluating, $0 = f(x) = c\, x_n^{N^*} + \ldots$, so after dividing by c, x_n is integral over A'. Then $x_i = x'_i + x_n^{e^i} \in A'[x_n]$, so $A = A'[x_n]$, which is integral over A' (5.2, 5.3). Now $A' \neq 0$ (it contains $1 \neq 0$) is generated by $n-1$ elements, so by induction there are $y_1, \ldots, y_r \in A'$ algebraically independent with A' integral over $k[y]$; by transitivity (5.4) A is integral over $k[y]$. □

The linear version when k is infinite. Suppose k is infinite and $f \neq 0$ as above, with F its homogeneous part of highest degree d. The polynomial $F(t_1, \ldots, t_{n-1}, 1) \in k[t_1, \ldots, t_{n-1}]$ is non-zero (dehomogenization of a non-zero homogeneous polynomial is non-zero) and k is infinite, so there are $\lambda_i \in k$ with $F(\lambda_1, \ldots, \lambda_{n-1}, 1) \neq 0$. With $x'_i = x_i - \lambda_i x_n$, the polynomial $f(t'_1 + \lambda_1 t_n, \ldots, t'_{n-1} + \lambda_{n-1} t_n, t_n)$ equals $F(\lambda, 1) t_n^d +$ (terms of degree $< d$ in t_n), whose leading coefficient is a non-zero element of k. As before x_n is integral over $A' = k[x'_1, \ldots, x'_{n-1}]$ and A is integral over A'. By induction the y_i can be taken to be k-linear combinations of $x_1, \ldots, x_n$ (each x'_i is one).

Geometric interpretation. Let k be algebraically closed (so infinite), $X \subseteq k^n$ an affine variety with coordinate ring $A = k[t_1, \ldots, t_n]/I(X) \neq 0$ generated by the coordinate functions $\xi_1, \ldots, \xi_n$. By the above, there are $y_i = \sum_j c_{ij} \xi_j$ $(1 \le i \le r)$ algebraically independent with A integral over $k[y_1, \ldots, y_r]$. Let $\pi : k^n \to k^r$, $\pi(x) = (\ell_1(x), \ldots, \ell_r(x))$ with $\ell_i = \sum_j c_{ij} t_j$. The ℓ_i are linearly independent forms (a linear relation among them would give an algebraic relation among the y_i), so π is a surjective linear map. We claim $\pi(X) = k^r$. Let $a \in k^r$. The homomorphism $k[y_1, \ldots, y_r] \to k$, $y_i \mapsto a_i$, extends to a homomorphism $g : A \to k$ by Exercise 2 (applied to the subring $k[y] \subseteq A$ over which A is integral, with $\Omega = k$). It restricts to the identity on k. Put $x_0 = (g(\xi_1), \ldots, g(\xi_n))$. For a polynomial $h \in I(X)$, $h(x_0) = g(h(\xi)) = g(0) = 0$; so $x_0 \in X$ (as X is defined by equations lying in $I(X)$). And $\ell_i(x_0) = g(y_i) = a_i$, so $\pi(x_0) = a$. Composing π with an embedding $\iota : k^r \to k^n$ identifies the image with a linear subspace $L = \iota(k^r)$ of dimension r and gives a linear map $k^n \to L$ mapping X onto L. □

Exercise 17

> **Problem.** (*Nullstellensatz, weak form.*) Let X be an affine algebraic variety in k^n, where k is an algebraically closed field, and let $I(X)$ be the ideal of X in the polynomial ring $k[t_1, \ldots, t_n]$ (Chapter 1, Exercise 27). If $I(X) \neq (1)$ then X is not empty. [Let $A = k[t_1, \ldots, t_n]/I(X)$ be the coordinate ring of X. Then $A \neq 0$, hence by Exercise 16 there exists a linear subspace L of dimension ≥ 0 in k^n and a mapping of X onto L. Hence $X \neq \emptyset$.]
>
> Deduce that every maximal ideal in the ring $k[t_1, \ldots, t_n]$ is of the form $(t_1 - a_1, \ldots, t_n - a_n)$ where $a_i \in k$.

Solution. *Remark.* Read literally, "$I(X) \neq (1) \Rightarrow X \neq \emptyset$" is immediate ($X = \emptyset$ gives $1 \in I(X)$). What is meant is the weak Nullstellensatz: *if $\mathfrak{a} \neq (1)$ is an ideal of $k[t_1, \ldots, t_n]$, then its zero set $V(\mathfrak{a}) \subseteq k^n$ is non-empty* (apply it to the ideal generated by the equations f_α defining X; one has $\mathfrak{a} \subseteq I(X)$, so $I(X) \neq (1) \Rightarrow \mathfrak{a} \neq (1)$). I prove this.

Let $A = k[t_1, \ldots, t_n]/\mathfrak{a} \neq 0$, generated by the images ξ_i of the t_i. By Exercise 16 there are $y_1, \ldots, y_r \in A$ algebraically independent with A integral over $k[y]$. Send $y_i \mapsto 0$: the homomorphism $k[y] \to k$ extends to a homomorphism $g : A \to k$ (Exercise 2, $\Omega = k$), which is the identity on k. Put $x_0 = (g(\xi_1), \ldots, g(\xi_n))$. For $h \in \mathfrak{a}$, $h(x_0) = g(h(\xi)) = g(0) = 0$; so $x_0 \in V(\mathfrak{a})$, which is therefore non-empty.

Maximal ideals. Let $\mathfrak{m}$ be a maximal ideal of $k[t]$; then $\mathfrak{m} \neq (1)$, so there is $a = (a_1, \ldots, a_n) \in V(\mathfrak{m})$. Every $h \in \mathfrak{m}$ vanishes at a, so $\mathfrak{m} \subseteq \mathrm{Ker}(\mathrm{ev}_a) = (t_1 - a_1, \ldots, t_n - a_n)$, which is a proper ideal (the kernel of the surjection $\mathrm{ev}_a : k[t] \to k$). By maximality $\mathfrak{m} = (t_1 - a_1, \ldots, t_n - a_n)$. □

Exercise 18

> **Problem.** Let k be a field and let B be a finitely generated k-algebra. Suppose that B is a field. Then B is a finite algebraic extension of k. (This is another version of Hilbert's Nullstellensatz. The following proof is due to Zariski. For other proofs, see (5.24), (7.9).)
>
> [Let $x_1, \ldots, x_n$ generate B as a k-algebra. The proof is by induction on n. If $n = 1$ the result is clearly true, so assume $n > 1$. Let $A = k[x_1]$ and let $K = k(x_1)$ be the field of fractions of A. By the inductive hypothesis, B is a finite algebraic extension of K, hence each of $x_2, \ldots, x_n$ satisfies a monic polynomial equation with coefficients in K, i.e. coefficients of the form a/b where a and b are in A. If f is the product of the denominators of all these coefficients, then each of $x_2, \ldots, x_n$ is integral over A_f. Hence B and therefore K is integral over A_f.
>
> Suppose x_1 is transcendental over k. Then A is integrally closed, because it is a unique factorization domain. Hence A_f is integrally closed (5.12), and therefore $A_f = K$, which is clearly absurd. Hence x_1 is algebraic over k, hence K (and therefore B) is a finite extension of k.]

Solution. Induct on the number n of generators $x_1, \ldots, x_n$ of the k-algebra B, which is a field.

$n = 1$. $B = k[x_1]$ is a field. If x_1 were transcendental, B would be a polynomial ring, not a field. So x_1 is algebraic and $B = k[x_1]$ is a finite extension of k.

$n > 1$. Let $A = k[x_1]$, $K = \mathrm{Frac}(A) = k(x_1) \subseteq B$ (as B is a field). B is generated over K by $x_2, \ldots, x_n$, $n - 1$ elements; by the inductive hypothesis B is a finite algebraic extension of K. So each x_i ($i \geq 2$) satisfies a monic polynomial over K, whose coefficients are quotients of elements of A. Let $f \in A - \{0\}$ be the product of all the denominators; then $x_2, \ldots, x_n$ are integral over A_f. Since $A \subseteq A_f \subseteq K \subseteq B$ and $B = A[x_2, \ldots, x_n]$, we have $B = A_f[x_2, \ldots, x_n]$, which is integral over A_f (5.2, 5.3). In particular $K \subseteq B$ is integral over A_f.

Suppose x_1 were transcendental over k. Then $A = k[x_1]$ is a UFD, hence integrally closed, so A_f is integrally closed (5.12/5.13), and $K = \mathrm{Frac}(A_f)$ integral over A_f would force $A_f = K$. But this is false: the polynomial $g = 1 + x_1 f$ has degree ≥ 1 and is coprime to f (it is $\equiv 1$ modulo every divisor of f), so $g \nmid f^m$ for any m, and $1/g \in K$ is not of the form $h/f^m \in A_f$. This contradiction shows that x_1 is algebraic over k. Then $K = k(x_1) = k[x_1]$ is a finite extension of k, and B is finite over K, hence finite (and algebraic) over k. □

Exercise 19

> **Problem.** Deduce the result of Exercise 17 from Exercise 18.

Solution. Let k be algebraically closed and $\mathfrak{a} \neq (1)$ an ideal of $k[t_1, \ldots, t_n]$ (the weak Nullstellensatz in the form proved in Exercise 17). Choose a maximal ideal $\mathfrak{m} \supseteq \mathfrak{a}$. The field $B = k[t]/\mathfrak{m}$ is a finitely generated k-algebra, so by Exercise 18 it is a finite algebraic extension of k, hence equal to k as k is algebraically closed (an element algebraic over k has a linear minimal polynomial). So each t_i has image some $a_i \in k$, i.e. $t_i - a_i \in \mathfrak{m}$ and $(t_1 - a_1, \ldots, t_n - a_n) \subseteq \mathfrak{m}$. The former is a maximal ideal (kernel of evaluation, a surjection onto k), so $\mathfrak{m} = (t_1 - a_1, \ldots, t_n - a_n)$. Then every $h \in \mathfrak{a} \subseteq \mathfrak{m}$ vanishes at $a = (a_1, \ldots, a_n)$, so $a \in V(\mathfrak{a}) \neq \emptyset$. Taking $\mathfrak{a}$ to be the ideal generated by the equations defining a variety X with $I(X) \neq (1)$ gives $X \neq \emptyset$. (This also gives again that every maximal ideal of $k[t]$ has the form $(t_1 - a_1, \ldots, t_n - a_n)$.) $\square$

Exercise 20

> **Problem.** Let A be a subring of an integral domain B such that B is finitely generated over A. Show that there exists $s \neq 0$ in A and elements $y_1, \ldots, y_n$ in B, algebraically independent over A and such that B_s is integral over B'_s, where $B' = A[y_1, \ldots, y_n]$. [Let $S = A - \{0\}$ and let $K = S^{-1}A$, the field of fractions of A. Then $S^{-1}B$ is a finitely generated K-algebra and therefore by the normalization lemma (Exercise 16) there exist $x_1, \ldots, x_n$ in $S^{-1}B$, algebraically independent over K and such that $S^{-1}B$ is integral over $K[x_1, \ldots, x_n]$. Let $z_1, \ldots, z_m$ generate B as an A-algebra. Then each z_j (regarded as an element of $S^{-1}B$) is integral over $K[x_1, \ldots, x_n]$. By writing an equation of integral dependence for each z_j, show that there exists $s \in S$ such that $x_i = y_i/s$ $(1 \leq i \leq n)$ with $y_i \in B$, and such that each sz_j is integral over B'_s. Deduce that this s satisfies the conditions stated.]

Solution. Let $S = A - \{0\}$, $K = S^{-1}A$. Since B is a domain containing A, $S^{-1}B \neq 0$ is a domain containing B; it is generated as a K-algebra by the generators $z_1, \ldots, z_m$ of B over A. By Exercise 16 there are $x_1, \ldots, x_n \in S^{-1}B$ algebraically independent over K with $S^{-1}B$ integral over $K[x_1, \ldots, x_n]$.

Write $x_i = b_i/t_i$ with $b_i \in B$, $t_i \in S$. Each z_j is integral over $K[x]$: $z_j^N + g_{j1}z_j^{N-1} + \cdots + g_{jN} = 0$ with $g_{jl} \in K[x_1, \ldots, x_n]$ (take N common to all j by multiplying by powers of z_j if necessary). The finitely many coefficients (in K) of all the polynomials g_{jl} have a common denominator $s_0 \in S$. Put $s = s_0 t_1 \cdots t_n \in S$ and $y_i = sx_i = b_i\, s_0 \prod_{k \neq i} t_k \in B$. Then all coefficients lie in A_s, and $x_i = y_i/s$, so $g_{jl} \in A_s[x_1, \ldots, x_n] = A_s[y_1, \ldots, y_n]$. Since $y_i = sx_i$ with $s \in K^\times$, the y_i are algebraically independent over K, hence over A. The equation for z_j holds in $S^{-1}B \supseteq B_s$ and has coefficients in $A_s[y] = B'_s$, so each z_j is integral over B'_s. Now $B_s = A_s[z_1, \ldots, z_m] = B'_s[z_1, \ldots, z_m]$ (as $B = A[z]$ and $B' \subseteq B$), so B_s is integral over B'_s by 5.2, 5.3. $\square$

Exercise 21

> **Problem.** Let A, B be as in Exercise 20. Show that there exists $s \neq 0$ in A such that, if Ω is an algebraically closed field and $f : A \to \Omega$ is a homomorphism for which $f(s) \neq 0$, then f can be extended to a homomorphism $B \to \Omega$. [With the notation of Exercise 20, f can be extended first of all to B', for example by mapping each y_i to 0; then to B'_s (because $f(s) \neq 0$), and finally to B_s (by Exercise 2, because B_s is integral over B'_s).]

Solution. Take s and $y_1, \ldots, y_n$ as in Exercise 20. Let $f : A \to \Omega$ with $f(s) \neq 0$. Since $f(s)$ is a unit in Ω, f extends to A_s by $f(a/s^k) = f(a)f(s)^{-k}$. As the y_i are algebraically independent

over A, $A[y]$ and $A_s[y] = B'_s$ are polynomial rings, so the homomorphism $A_s \to \Omega$ extends to $B'_s = A_s[y_1, \ldots, y_n] \to \Omega$ by $y_i \mapsto 0$. Since B_s is integral over B'_s, Exercise 2 extends this to a homomorphism $B_s \to \Omega$. Its restriction to $B \subseteq B_s$ extends f. □

Exercise 22

Problem. Let A, B be as in Exercise 20. If the Jacobson radical of A is zero, then so is the Jacobson radical of B. [Let $v \neq 0$ be an element of B. We have to show that there is a maximal ideal of B which does not contain v. By applying Exercise 21 to the ring B_v and its subring A, we obtain an element $s \neq 0$ in A. Let $\mathfrak{m}$ be a maximal ideal of A such that $s \notin \mathfrak{m}$, and let $k = A/\mathfrak{m}$. Then the canonical mapping $A \to k$ extends to a homomorphism g of B_v into an algebraic closure Ω of k. Show that $g(v) \neq 0$ and that $\mathrm{Ker}(g) \cap B$ is a maximal ideal of B.]

Solution. Let $v \in B$, $v \neq 0$. Then B_v is a domain containing A, finitely generated as an A-algebra (by the generators of B and $1/v$). By Exercise 21 applied to $A \subseteq B_v$ there is $s \neq 0$ in A with the stated extension property. Since $\mathfrak{R}(A) = 0$ and $s \neq 0$, s is not in some maximal ideal $\mathfrak{m}$ of A. Let $k = A/\mathfrak{m}$, Ω an algebraic closure of k, and $f : A \to k \to \Omega$ the canonical map; $f(s) \neq 0$, and $\mathrm{Ker}\, f = \mathfrak{m}$. By Exercise 21, f extends to $g : B_v \to \Omega$. Since v is a unit of B_v, $g(v)g(v^{-1}) = 1$, so $g(v) \neq 0$.

Let $g_0 = g|_B$ and $\mathfrak{n} = \mathrm{Ker}\, g_0$. The image $g_0(B)$ is a subring of Ω containing $g(A) = k$. It is generated as a k-algebra by the images of generators of B over A, which are algebraic over k (all elements of Ω are), hence integral over the field k. So $g_0(B) \supseteq k$ is an integral domain integral over the field k, hence a field (5.7). Thus $B/\mathfrak{n} \cong g_0(B)$ is a field and $\mathfrak{n}$ is a maximal ideal of B; and $v \notin \mathfrak{n}$ since $g(v) \neq 0$. As $v \neq 0$ was arbitrary, $\mathfrak{R}(B) = 0$. □

Exercise 23

Problem. Let A be a ring. Show that the following are equivalent:

(i) Every prime ideal in A is an intersection of maximal ideals.

(ii) In every homomorphic image of A the nilradical is equal to the Jacobson radical.

(iii) Every prime ideal in A which is not maximal is equal to the intersection of the prime ideals which contain it strictly.

[The only hard part is iii)⇒ii). Suppose ii) false; then there is a prime ideal which is not an intersection of maximal ideals. Passing to the quotient ring, we may assume that A is an integral domain whose Jacobson radical $\mathfrak{R}$ is not zero. Let f be a non-zero element of $\mathfrak{R}$. Then $A_f \neq 0$, hence A_f has a maximal ideal, whose contraction in A is a prime ideal $\mathfrak{p}$ such that $f \notin \mathfrak{p}$, and which is maximal with respect to this property. Then $\mathfrak{p}$ is not maximal and is not equal to the intersection of the prime ideals strictly containing $\mathfrak{p}$.]

A ring A with the three equivalent properties above is called a *Jacobson ring*.

Solution. Recall that the primes of a quotient $A/\mathfrak{a}$ correspond to the primes of A containing $\mathfrak{a}$, maximal ones to maximal ones, and $\mathfrak{N} \subseteq \mathfrak{R}$ always.

(i)⇒(ii). Property (i) passes to quotients: a prime of $A/\mathfrak{a}$ is $\mathfrak{p}/\mathfrak{a}$ with $\mathfrak{p} \supseteq \mathfrak{a}$ prime in A, $\mathfrak{p} = \bigcap \mathfrak{m}_i$ with $\mathfrak{m}_i \supseteq \mathfrak{p} \supseteq \mathfrak{a}$ maximal, so $\mathfrak{p}/\mathfrak{a} = \bigcap(\mathfrak{m}_i/\mathfrak{a})$. So it suffices to show $\mathfrak{N} = \mathfrak{R}$ in A itself. Each prime of A is an intersection of maximal ideals, hence contains $\mathfrak{R}$; so $\mathfrak{N} = \bigcap \mathfrak{p} \supseteq \mathfrak{R}$ and $\mathfrak{N} = \mathfrak{R}$.

(ii)⇒(i). Let $\mathfrak{p}$ be prime. In $A/\mathfrak{p}$, $\mathfrak{R} = \mathfrak{N} = 0$ (a domain). So $\mathfrak{p} = \bigcap$ of the maximal ideals containing $\mathfrak{p}$.

(ii)⇒(iii). Let $\mathfrak{p}$ be a prime which is not maximal and $f \in A$ lie in every prime strictly containing $\mathfrak{p}$. In the domain $D = A/\mathfrak{p}$, which is not a field, every maximal ideal is non-zero, hence corresponds to a prime strictly containing $\mathfrak{p}$, so the image $\bar{f}$ lies in all maximal ideals of D: $\bar{f} \in \mathfrak{R}(D) = \mathfrak{N}(D) = 0$ by (ii). So $f \in \mathfrak{p}$. Hence $\mathfrak{p}$ is the intersection of the primes strictly containing it (and there is at least one, a maximal ideal).

(iii)⇒(ii). Suppose (ii) fails: there is an ideal $\mathfrak{a}$ and $x \in \mathfrak{R}(A/\mathfrak{a})$ not nilpotent. Then some prime of $A/\mathfrak{a}$ does not contain x; passing to the quotient by it, we get a domain $D = A/\mathfrak{p}_0$ and a non-zero element $\bar{x} \in \mathfrak{R}(D)$ (the image of $\mathfrak{R}(R)$ in a quotient R/I lies in $\mathfrak{R}(R/I)$, because maximal ideals of R/I pull back to maximal ideals of R). Take $f \neq 0$ in $\mathfrak{R}(D)$; $D_f \neq 0$ has a maximal ideal, whose contraction $\mathfrak{p}$ in D is a prime with $f \notin \mathfrak{p}$, maximal with respect to this property (primes of D_f correspond to primes of D not containing f). $\mathfrak{p}$ is not maximal (a maximal ideal of D contains $f \in \mathfrak{R}(D)$), and every prime strictly containing $\mathfrak{p}$ contains f (by the maximality), so the intersection of the primes strictly containing $\mathfrak{p}$ contains $f \notin \mathfrak{p}$ and differs from $\mathfrak{p}$. Pulling $\mathfrak{p}$ back to A gives a prime of A violating (iii) (primes of D correspond to primes of A containing $\mathfrak{p}_0$, preserving inclusions). □

Exercise 24

> **Problem.** Let A be a Jacobson ring (Exercise 23) and B an A-algebra. Show that if B is either (i) integral over A or (ii) finitely generated as an A-algebra, then B is a Jacobson ring. [Use Exercise 22 for (ii).]
>
> In particular, every finitely generated ring, and every finitely generated algebra over a field, is a Jacobson ring.

Solution. Let $f : A \to B$ be the structure map and let $\mathfrak{q}$ be a prime of B, $\mathfrak{p} = \mathfrak{q}^c$. By Exercise 23(i)⇔(ii) it suffices to show that $\mathfrak{q}$ is an intersection of maximal ideals, i.e. that $\mathfrak{R}(B/\mathfrak{q}) = 0$. Now $A' = A/\mathfrak{p} \hookrightarrow B/\mathfrak{q} = B'$ (the kernel of $A \to B/\mathfrak{q}$ is $\mathfrak{q}^c = \mathfrak{p}$), and A' is a Jacobson domain (a quotient of a Jacobson ring is Jacobson, see the proof of (i)⇒(ii) in Exercise 23), so 0 is an intersection of maximal ideals of A', i.e. $\mathfrak{R}(A') = 0$.

(i) If B is integral over A, then B' is integral over A' (5.6), and $\mathfrak{R}(A') = \mathfrak{R}(B') \cap A'$ (Exercise 5(ii)). Hence $\mathfrak{R}(B') \cap A' = 0$. Now B' is a domain integral over the domain A'; suppose $\mathfrak{R}(B') \neq 0$ and take a non-zero $b \in \mathfrak{R}(B')$; it satisfies a monic equation $b^m + a_1 b^{m-1} + \cdots + a_m = 0$ of least degree; then $a_m \neq 0$ (else we divide by b in the domain and reduce the degree), and $a_m = -b(b^{m-1} + \cdots + a_{m-1}) \in \mathfrak{R}(B') \cap A'$ (as $\mathfrak{R}(B')$ is an ideal). This contradicts $\mathfrak{R}(B') \cap A' = 0$. So $\mathfrak{R}(B') = 0$.

(ii) If B is a finitely generated A-algebra, B' is a domain finitely generated over $A' \subseteq B'$ with $\mathfrak{R}(A') = 0$. By Exercise 22, $\mathfrak{R}(B') = 0$.

In both cases every prime of B is an intersection of maximal ideals, so B is Jacobson. Finally $\mathbf{Z}$ is a Jacobson ring (its primes are (0), which is the intersection of the infinitely many maximal

ideals (p), and the maximal ideals (p)), and a field is Jacobson (its only prime is maximal). A finitely generated ring is a finitely generated $\mathbf{Z}$-algebra, and a finitely generated algebra over a field is a finitely generated algebra over that field, so both are Jacobson by (ii). □

Exercise 25

Problem. Let A be a ring. Show that the following are equivalent:

(i) A is a Jacobson ring.

(ii) Every finitely generated A-algebra B which is a field is finite over A.

[(i)⇒(ii). Reduce to the case where A is a subring of B, and use Exercise 21. If $s \in A$ is as in Exercise 21, then there exists a maximal ideal $\mathfrak{m}$ of A not containing s, and the homomorphism $A \to A/\mathfrak{m} = k$ extends to a homomorphism g of B into the algebraic closure of k. Since B is a field, g is injective, and $g(B)$ is algebraic over k, hence finite over k.

(ii)⇒(i). Use criterion iii) of Exercise 23. Let $\mathfrak{p}$ be a prime ideal of A which is not maximal, and let $B = A/\mathfrak{p}$. Let f be a non-zero element of B. Then B_f is a finitely generated A-algebra. If it is a field it is finite over A (by ii)) hence integral over B and therefore B is a field by (5.7). Hence B_f is not a field and therefore has a non-zero prime ideal, whose contraction in B is a non-zero ideal $\mathfrak{p}'$ such that $f \notin \mathfrak{p}'$.]

Solution. (i)⇒(ii). Let B be a finitely generated A-algebra which is a field, $\mathfrak{p} = \mathrm{Ker}(A \to B)$, a prime ideal. Then $A' = A/\mathfrak{p} \subseteq B$ is a Jacobson domain (a quotient of a Jacobson ring), so $\mathfrak{R}(A') = 0$; and B is a finitely generated A'-algebra, and finiteness over A is finiteness over A'. Apply Exercise 21 to $A' \subseteq B$: we get $s \neq 0$ in A'. Since $\mathfrak{R}(A') = 0$ there is a maximal ideal $\mathfrak{m}'$ of A' with $s \notin \mathfrak{m}'$. Let $k = A'/\mathfrak{m}'$, Ω an algebraic closure of k, and $f : A' \to k \to \Omega$ the canonical map; $f(s) \neq 0$. By Exercise 21, f extends to $g : B \to \Omega$. As B is a field, g is injective ($g(1) = 1 \neq 0$). Then $g|_{A'}$ is injective, but its kernel is $\mathfrak{m}'$, so $\mathfrak{m}' = 0$ and $A' = k$ is a field. Now $g(B)$ is generated over $g(A') = k$ by finitely many elements of Ω, which are algebraic over k, so $g(B)$ is a finite extension of k. Therefore $B \cong g(B)$ is finite over A', hence finite over A.

(ii)⇒(i). We verify Exercise 23(iii). Let $\mathfrak{p}$ be a prime of A which is not maximal, $B = A/\mathfrak{p}$, and $0 \neq f \in B$. $B_f = B[1/f]$ is a finitely generated A-algebra. If B_f were a field, then by (ii) it would be finite over A, hence integral over the image $B = A/\mathfrak{p}$ of A in it (note $B \subseteq B_f$ as B is a domain); then B would be a field by 5.7, contradicting that $\mathfrak{p}$ is not maximal. So the domain B_f is not a field, and has a non-zero maximal ideal $\mathfrak{Q}$. Let $\mathfrak{p}' = \mathfrak{Q} \cap B$, a prime of B. Then $f \notin \mathfrak{p}'$ (as f is a unit of B_f), and $\mathfrak{p}' \neq 0$: a non-zero element $b/f^k \in \mathfrak{Q}$ gives $b \in \mathfrak{Q} \cap B$, $b \neq 0$. So the prime of A corresponding to $\mathfrak{p}'$ strictly contains $\mathfrak{p}$ and does not contain (a lift of) f. As $f \in B - \{0\}$ was arbitrary, $\mathfrak{p}$ equals the intersection of the primes strictly containing it. Hence A is Jacobson by Exercise 23. □

Exercise 26

Problem. Let X be a topological space. A subset of X is *locally closed* if it is the intersection of an open set and a closed set, or equivalently if it is open in its closure.

The following conditions on a subset X_0 of X are equivalent:

(1) Every non-empty locally closed subset of X meets X_0;

(2) For every closed set E in X we have $\overline{E \cap X_0} = E$;

(3) The mapping $U \mapsto U \cap X_0$ of the collection of open sets of X onto the collection of open sets of X_0 is bijective.

A subset X_0 satisfying these conditions is said to be *very dense* in X.

If A is a ring, show that the following are equivalent:

(i) A is a Jacobson ring;

(ii) The set of maximal ideals of A is very dense in $\mathrm{Spec}(A)$;

(iii) Every locally closed subset of $\mathrm{Spec}(A)$ consisting of a single point is closed.

[(ii) and (iii) are geometrical formulations of conditions ii) and iii) of Exercise 23.]

Solution. The three conditions on X_0 are equivalent.

(1)⇒(2): Let E be closed and $F = \overline{E \cap X_0} \subseteq E$. If $F \neq E$, then $E - F = E \cap (X - F)$ is a non-empty locally closed set, so by (1) it contains a point of X_0. This point lies in $E \cap X_0 \subseteq F$, a contradiction.

(2)⇒(3): The map is onto the open sets of X_0 by the definition of the subspace topology. If $U \cap X_0 = U' \cap X_0$, put $E = X - U$, $E' = X - U'$; then $E \cap X_0 = X_0 - U \cap X_0 = E' \cap X_0$, so by (2) $E = \overline{E \cap X_0} = \overline{E' \cap X_0} = E'$, and $U = U'$.

(3)⇒(1): Let $L = U \cap E \neq \emptyset$ (U open, E closed) with $L \cap X_0 = \emptyset$. Then $U \cap X_0 = (U \cap (X - E)) \cap X_0$, and $U \cap (X - E)$ is open, so by injectivity in (3) $U = U - E$, i.e. $U \cap E = \emptyset$, contradicting $L \neq \emptyset$.

Spectra. Let $X = \mathrm{Spec}\, A$, $X_0 = \mathrm{Max}\, A$. Recall that the closure of $S \subseteq X$ is $V(\bigcap_{\mathfrak{p} \in S} \mathfrak{p})$.

(i)⇔(ii). Condition (2) for X_0 and the closed set $E = V(\mathfrak{a})$ says $V\big(\bigcap_{\mathfrak{m} \supseteq \mathfrak{a}} \mathfrak{m}\big) = V(\mathfrak{a})$, which holds iff $\bigcap_{\mathfrak{m} \supseteq \mathfrak{a}} \mathfrak{m} = r(\mathfrak{a})$ (the left side is a radical ideal, and $V(\mathfrak{b}) = V(\mathfrak{a})$ iff $r(\mathfrak{b}) = r(\mathfrak{a})$, Ch. 1, Ex. 15). Writing this in $A/\mathfrak{a}$, it says $\mathfrak{R}(A/\mathfrak{a}) = \mathfrak{N}(A/\mathfrak{a})$. So (ii) holds iff this is true for all ideals $\mathfrak{a}$, i.e. iff Exercise 23(ii) holds, i.e. iff A is Jacobson.

(i)⇒(iii). Let $\{\mathfrak{p}\} = U \cap V(\mathfrak{a})$ be locally closed (U open). Since $U \ni \mathfrak{p}$, $U \supseteq X_f$ for some $f \notin \mathfrak{p}$ (basic opens, Ch. 1, Ex. 17), and then $\{\mathfrak{p}\} = X_f \cap V(\mathfrak{a})$ (it contains $\mathfrak{p}$ and is contained in $U \cap V(\mathfrak{a})$). Suppose $\mathfrak{p}$ is not maximal. Every prime strictly containing $\mathfrak{p}$ lies in $V(\mathfrak{a})$ (as $\mathfrak{a} \subseteq \mathfrak{p}$) and is not in X_f (else it would be an element of $\{\mathfrak{p}\}$), so it contains f. Then $f \in \bigcap\{\mathfrak{p}' \supsetneq \mathfrak{p}\} = \mathfrak{p}$ by Exercise 23(iii), a contradiction. So $\mathfrak{p}$ is maximal, and $\{\mathfrak{p}\}$ is closed.

(iii)⇒(i). We verify Exercise 23(iii). Let $\mathfrak{p}$ be non-maximal and suppose some $f \notin \mathfrak{p}$ lies in every prime strictly containing $\mathfrak{p}$. Then $V(\mathfrak{p}) \cap X_f = \{\mathfrak{p}\}$, a locally closed single point, which is closed by (iii); so $\mathfrak{p}$ is maximal, contradiction. Hence $\mathfrak{p}$ is the intersection of the primes strictly containing it. □

Exercise 27

Problem. Let A, B be two local rings. B is said to *dominate* A if A is a subring of B and the maximal ideal $\mathfrak{m}$ of A is contained in the maximal ideal $\mathfrak{n}$ of B (or, equivalently, if $\mathfrak{m} = \mathfrak{n} \cap A$). Let K be a field and let Σ be the set of all local subrings of K. If Σ is ordered by the relation of domination, show that Σ has maximal elements and that $A \in \Sigma$ is maximal if and only if A is a valuation ring of K. [Use (5.21).]

Solution. Domination is a partial order on Σ (reflexive and transitive clearly; if A dominates B and B dominates A then $A = B$). ($\Sigma \ni K$ is non-empty.)

Existence of maximal elements. Let (A_α) be a chain in Σ, $C = \bigcup A_\alpha$ (a subring of K), and $\mathfrak{m}_C = \bigcup \mathfrak{m}_\alpha$. If A_β dominates A_α then $\mathfrak{m}_\beta \cap A_\alpha = \mathfrak{m}_\alpha$, so $\mathfrak{m}_\alpha \subseteq \mathfrak{m}_\beta$ along the chain and $\mathfrak{m}_C$ is an ideal of C; it is proper ($1 \notin \mathfrak{m}_\alpha$ for all α). An element $x \in C - \mathfrak{m}_C$ lies in some A_α and not in $\mathfrak{m}_\alpha$, hence is a unit of $A_\alpha \subseteq C$. So C is local with maximal ideal $\mathfrak{m}_C$; and $\mathfrak{m}_C \cap A_\alpha = \mathfrak{m}_\alpha$ (an element of A_α in $\mathfrak{m}_\beta$, $\beta \geq \alpha$, lies in $\mathfrak{m}_\beta \cap A_\alpha = \mathfrak{m}_\alpha$). So C dominates each A_α. By Zorn's lemma Σ has maximal elements.

Maximal $\Rightarrow$ valuation ring. Let $A \in \Sigma$ be maximal, $\mathfrak{m}$ its maximal ideal, Ω an algebraic closure of $A/\mathfrak{m}$ and $g : A \to A/\mathfrak{m} \to \Omega$, so $\operatorname{Ker} g = \mathfrak{m}$. We show (A, g) is maximal in the set of pairs of 5.21. Let (B, h) be a pair with $A \subseteq B \subseteq K$ and $h|_A = g$. The kernel $\mathfrak{P} = \operatorname{Ker} h$ is a prime ideal of B (the image lies in a field), $0 \in \mathfrak{P}$, so $B' = B_{\mathfrak{P}}$ is a local subring of K with maximal ideal $\mathfrak{P}B'$, and $\mathfrak{P} \cap A = \operatorname{Ker} g = \mathfrak{m}$. So $\mathfrak{m} = \mathfrak{P}B' \cap A \subseteq \mathfrak{P}B'$, i.e. B' dominates A. By maximality of A, $B' = A$, so $B \subseteq B' = A$, whence $B = A$ and $h = g$. So (A, g) is a maximal pair and, by Theorem 5.21, A is a valuation ring of K.

Valuation ring $\Rightarrow$ maximal. Let A be a valuation ring of K and let $B \in \Sigma$ dominate A, with maximal ideals $\mathfrak{m}_A \subseteq \mathfrak{m}_B$. Let $x \in B$. If $x \notin A$ then $x^{-1} \in A$ (valuation ring), and x^{-1} is not a unit of A (else $x \in A$), so $x^{-1} \in \mathfrak{m}_A \subseteq \mathfrak{m}_B$; then $1 = x \cdot x^{-1} \in \mathfrak{m}_B$ (as $x \in B$), absurd. So $B \subseteq A$, and since $A \subseteq B$, $A = B$. Hence A is maximal. □

Exercise 28

Problem. Let A be an integral domain, K its field of fractions. Show that the following are equivalent:

(1) A is a valuation ring of K;

(2) If $\mathfrak{a}, \mathfrak{b}$ are any two ideals of A, then either $\mathfrak{a} \subseteq \mathfrak{b}$ or $\mathfrak{b} \subseteq \mathfrak{a}$.

Deduce that if A is a valuation ring and $\mathfrak{p}$ is a prime ideal of A, then $A_{\mathfrak{p}}$ and $A/\mathfrak{p}$ are valuation rings of their fields of fractions.

Solution. (1)⇒(2). Suppose $\mathfrak{a} \not\subseteq \mathfrak{b}$ and pick $x \in \mathfrak{a} - \mathfrak{b}$ (so $x \neq 0$). Let $y \in \mathfrak{b}$, $y \neq 0$. Either $y/x \in A$, giving $y = (y/x)x \in \mathfrak{a}$; or $x/y \in A$, giving $x \in yA \subseteq \mathfrak{b}$, a contradiction. So $\mathfrak{b} \subseteq \mathfrak{a}$.

(2)⇒(1). Let $0 \neq x = a/b \in K$ with $a, b \in A - \{0\}$. By (2), either $(a) \subseteq (b)$, so $a = bc$ and $x = c \in A$; or $(b) \subseteq (a)$, so $b = ad$ and $x^{-1} = d \in A$.

Consequences. Let A be a valuation ring and $\mathfrak{p}$ a prime. The ideals of $A_{\mathfrak{p}}$ are the extended ideals $\mathfrak{a}^e$ (3.11(i)); as the ideals $\mathfrak{a}$ of A are totally ordered by inclusion, so are the ideals of $A_{\mathfrak{p}}$. $A_{\mathfrak{p}}$ is a domain with fraction field K, so by (2)⇒(1) it is a valuation ring of K. The ideals of $A/\mathfrak{p}$ correspond to the ideals of A containing $\mathfrak{p}$, hence are totally ordered by inclusion, and $A/\mathfrak{p}$ is a domain; so by (2)⇒(1) it is a valuation ring of its fraction field. □

Exercise 29

Problem. Let A be a valuation ring of a field K. Show that every subring of K which contains A is a local ring of A.

Solution. Let B be a subring of K with $A \subseteq B$. By 5.18(ii), B is a valuation ring of K (for $x \neq 0$, $x \in A \subseteq B$ or $x^{-1} \in A \subseteq B$), and by 5.18(i) it is a local ring. (In fact $B = A_{\mathfrak{p}}$ with $\mathfrak{p} = \mathfrak{n} \cap A$, $\mathfrak{n}$ the maximal ideal of B: this is what "local ring of A" means. Indeed $A_{\mathfrak{p}} \subseteq B$ since the elements of $A - \mathfrak{p}$ are units of B (they lie in $B - \mathfrak{n}$); conversely, if $x \in B - A_{\mathfrak{p}}$ then $x^{-1} \in A$ (valuation ring), and $x^{-1} \in \mathfrak{p}$ (otherwise $x \in A_{\mathfrak{p}}$), so $x^{-1} \in \mathfrak{n}$, which contradicts $x \in B$, $x \cdot x^{-1} = 1$.) □

Exercise 30

Problem. Let A be a valuation ring of a field K. The group U of units of A is a subgroup of the multiplicative group K^* of K.

Let $\Gamma = K^*/U$. If $\xi, \eta \in \Gamma$ are represented by $x, y \in K^*$, define $\xi \geq \eta$ to mean $xy^{-1} \in A$. Show that this defines a total ordering on Γ which is compatible with the group structure (i.e., $\xi \geq \eta \Rightarrow \xi\omega \geq \eta\omega$ for all $\omega \in \Gamma$). In other words, Γ is a totally ordered abelian group. It is called the *value group* of A.

Let $v : K^* \to \Gamma$ be the canonical homomorphism. Show that $v(x+y) \geq \min(v(x), v(y))$ for all $x, y \in K^*$.

Solution. $U = A^\times$ is a subgroup of K^*, and K^* is abelian, so $\Gamma = K^*/U$ is an abelian group.

Well defined. Replace x, y by xu, yu' with $u, u' \in U$; then xy^{-1} changes by the unit uu'^{-1}, so $xy^{-1} \in A \iff (xu)(yu')^{-1} \in A$.

Total order. Reflexive: $xx^{-1} = 1 \in A$. Transitive: if $xy^{-1} \in A$ and $yz^{-1} \in A$ then $xz^{-1} = (xy^{-1})(yz^{-1}) \in A$. Antisymmetric: if $xy^{-1} \in A$ and $yx^{-1} \in A$ then xy^{-1} is a unit, so $\xi = \eta$. Total: for $x, y \in K^*$, $xy^{-1} \in A$ or $(xy^{-1})^{-1} = yx^{-1} \in A$ since A is a valuation ring.

Compatibility. If $\xi \geq \eta$ and ω is represented by w, then $(xw)(yw)^{-1} = xy^{-1} \in A$, so $\xi\omega \geq \eta\omega$.

The inequality. (Here $x + y \neq 0$ is understood, so that $v(x+y)$ is defined.) Say $v(x) \geq v(y)$, i.e. $x/y \in A$, so that $\min(v(x), v(y)) = v(y)$. Then $(x+y)/y = x/y + 1 \in A$, i.e. $v(x+y) \geq v(y) = \min(v(x), v(y))$. □

Exercise 31

Problem. Conversely, let Γ be a totally ordered abelian group (written *additively*), and let K be a field. A *valuation of K with values in Γ* is a mapping $v : K^* \to \Gamma$ such that

(1) $v(xy) = v(x) + v(y)$,

(2) $v(x+y) \geq \min(v(x), v(y))$,

for all $x, y \in K^*$. Show that the set of elements $x \in K^*$ such that $v(x) \geq 0$ is a valuation ring of K. This ring is called the *valuation ring* of v, and the subgroup $v(K^*)$ of Γ is the *value group* of v.

Thus the concepts of valuation ring and valuation are essentially equivalent.

Solution. (Condition (2) is understood whenever $x + y \neq 0$; if x or y is 0 in the sums below there is nothing to prove.) Let $A = \{0\} \cup \{x \in K^* : v(x) \geq 0\}$. From (1), $v(1) = v(1) + v(1)$, so

$v(1) = 0$; $2v(-1) = v(1) = 0$ and in a totally ordered group $\gamma > 0 \Rightarrow 2\gamma > 0$, $\gamma < 0 \Rightarrow 2\gamma < 0$, so $v(-1) = 0$ and $v(-x) = v(x)$; and $v(x^{-1}) = -v(x)$.

A is a subring. $1 \in A$; $-x \in A$ if $x \in A$; A is closed under products by (1); if $x, y \in A$ and $x + y \neq 0$ then $v(x + y) \geq \min(v(x), v(y)) \geq 0$ by (2).

Valuation ring. Let $x \in K^*$. If $v(x) \geq 0$ then $x \in A$; otherwise $v(x^{-1}) = -v(x) > 0$ and $x^{-1} \in A$. So for each $x \neq 0$ in K, $x \in A$ or $x^{-1} \in A$. In particular A is an integral domain with fraction field K, i.e. a valuation ring of K. □

Exercise 32

> **Problem.** Let Γ be a totally ordered abelian group. A subgroup Δ of Γ is *isolated* in Γ if, whenever $0 \leq \beta \leq \alpha$ and $\alpha \in \Delta$, we have $\beta \in \Delta$. Let A be a valuation ring of a field K, with value group Γ (Exercise 31). If $\mathfrak{p}$ is a prime ideal of A, show that $v(A - \mathfrak{p})$ is the set of elements ≥ 0 in an isolated subgroup Δ of Γ, and that the mapping so defined of $\mathrm{Spec}(A)$ into the set of isolated subgroups of Γ is bijective.
>
> If $\mathfrak{p}$ is a prime ideal of A, what are the value groups of the valuation rings $A/\mathfrak{p}$, $A_\mathfrak{p}$?

Solution. Write Γ additively with $v : K^* \to \Gamma$ surjective (Exercise 30 in additive notation) and $A = \{0\} \cup \{x : v(x) \geq 0\}$. Note: if $x, y \in A - \{0\}$ and $v(x) \leq v(y)$, then $y/x \in A$, so $y \in xA$.

The subgroup $\Delta_\mathfrak{p}$. Let $\mathfrak{p}$ be prime, $T = v(A - \mathfrak{p}) = \{v(x) : x \in A - \mathfrak{p}\} \subseteq \Gamma_{\geq 0}$ (note $0 \in \mathfrak{p}$, so only $x \neq 0$ occur). $0 = v(1) \in T$; T is closed under $+$ since $A - \mathfrak{p}$ is multiplicatively closed. *T is an initial segment of $\Gamma_{\geq 0}$:* if $0 \leq \beta \leq \alpha = v(x)$, $x \in A - \mathfrak{p}$, and $\beta = v(y)$ ($y \in A$ since $\beta \geq 0$), then $v(y) \leq v(x)$ gives $x \in yA$; if $y \in \mathfrak{p}$ we would have $x \in \mathfrak{p}$; so $y \in A - \mathfrak{p}$ and $\beta \in T$. Put $\Delta_\mathfrak{p} = T \cup (-T)$. It is a subgroup: $-\Delta = \Delta$, and for $\alpha, \beta \in T$, $\alpha - \beta \in [0, \alpha] \subseteq T$ if $\alpha \geq \beta$, and $\alpha - \beta \in -[0, \beta] \subseteq -T$ if $\alpha < \beta$. It is isolated: if $0 \leq \beta \leq \alpha \in \Delta_\mathfrak{p}$ then $\alpha \in T$ (as $\alpha \geq 0$ and $\Delta_\mathfrak{p} \cap \Gamma_{\geq 0} = T$), hence $\beta \in T$. And $\Delta_\mathfrak{p} \cap \Gamma_{\geq 0} = T = v(A - \mathfrak{p})$, as claimed.

Bijectivity. The map is $\mathfrak{p} \mapsto \Delta_\mathfrak{p}$. If $x \in A - \{0\}$ then $x \notin \mathfrak{p} \iff v(x) \in \Delta_\mathfrak{p}$ ("$\Rightarrow$" by definition; "$\Leftarrow$": $v(x) = v(y)$ with $y \in A - \mathfrak{p}$ gives $x/y \in A^\times$, so $x \in yA^\times \not\subseteq \mathfrak{p}$). So $\mathfrak{p} = \{0\} \cup \{x \in A - \{0\} : v(x) \notin \Delta_\mathfrak{p}\}$, whence the map is *injective*. For *surjectivity* let Δ be an isolated subgroup and put $\mathfrak{p}_\Delta = \{0\} \cup \{x \in A - \{0\} : v(x) \notin \Delta\}$. It is an ideal: if $x \in \mathfrak{p}_\Delta$ and $a \in A$, $a \neq 0$, then $0 \leq v(x) \leq v(ax)$, and $v(ax) \in \Delta$ would give $v(x) \in \Delta$ (isolated); and if $x, y \in \mathfrak{p}_\Delta$ with $x + y \neq 0$ and, say, $v(x) \leq v(y)$, then $v(x + y) \geq v(x)$, so $v(x + y) \in \Delta$ would give $v(x) \in \Delta$. It is proper ($v(1) = 0 \in \Delta$) and prime: if $x, y \in A - \mathfrak{p}_\Delta$ then $v(x), v(y) \in \Delta$, so $v(xy) \in \Delta$, so $xy \notin \mathfrak{p}_\Delta$. Finally $v(A - \mathfrak{p}_\Delta) = \{\alpha \in \Delta : \alpha \geq 0\}$ (every such α is $v(x)$ for some $x \in A$ since v is onto Γ), whence $\Delta_{\mathfrak{p}_\Delta} = \Delta$. Thus $\mathfrak{p} \mapsto \Delta_\mathfrak{p}$ is a bijection from $\mathrm{Spec}\, A$ onto the set of isolated subgroups, with inverse $\Delta \mapsto \mathfrak{p}_\Delta$; it reverses inclusions ($\mathfrak{p} \subseteq \mathfrak{q} \iff \Delta_\mathfrak{q} \subseteq \Delta_\mathfrak{p}$).

Value groups of $A/\mathfrak{p}$ and $A_\mathfrak{p}$.

$A_\mathfrak{p}$. $A_\mathfrak{p} = \{a/s : a \in A, s \in A - \mathfrak{p}\} = \{0\} \cup \{x \in K^* : v(x) \geq -\delta \text{ for some } \delta \in T\}$. Its units are the x with x and $x^{-1} \in A_\mathfrak{p}$, i.e. $-\delta \leq v(x) \leq \delta'$ with $\delta, \delta' \in T$, i.e. (by isolation) $v(x) \in \Delta_\mathfrak{p}$. So $U(A_\mathfrak{p}) = v^{-1}(\Delta_\mathfrak{p})$ and the value group of $A_\mathfrak{p}$ is $K^*/v^{-1}(\Delta_\mathfrak{p}) \cong \Gamma/\Delta_\mathfrak{p}$, with the order $\gamma + \Delta_\mathfrak{p} \geq 0$ iff $\gamma \geq \delta$ for some $\delta \in \Delta_\mathfrak{p}$ (well defined because $\Delta_\mathfrak{p}$ is isolated).

$A/\mathfrak{p}$. The fraction field of $A/\mathfrak{p}$ is the residue field $k(\mathfrak{p}) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ (Remark after 3.13). Now $\mathfrak{p}A_\mathfrak{p} = \{x : v(x) > \Delta_\mathfrak{p}\}$ (greater than every element of $\Delta_\mathfrak{p}$), and the units of $A_\mathfrak{p}$ are $\{x : v(x) \in$

$\Delta_{\mathfrak{p}}\}$, so v induces a map $w : k(\mathfrak{p})^* \to \Delta_{\mathfrak{p}}$, $w(\bar{x}) = v(x)$ for x a unit of $A_{\mathfrak{p}}$; it is well defined because $\bar{x} = \bar{x}'$ gives $x'/x \in 1 + \mathfrak{p}A_{\mathfrak{p}}$, which has value 0. It is a valuation (properties (1),(2) of Exercise 31 follow from those of v), surjective onto $\Delta_{\mathfrak{p}}$, and its valuation ring $\{\bar{x} : w(\bar{x}) \geq 0\} \cup \{0\}$ is the image of A, i.e. $A/\mathfrak{p}$. Hence *the value group of* $A/\mathfrak{p}$ *is* $\Delta_{\mathfrak{p}}$ (a subgroup of Γ), and *the value group of* $A_{\mathfrak{p}}$ *is* $\Gamma/\Delta_{\mathfrak{p}}$. □

Exercise 33

> **Problem.** Let Γ be a totally ordered abelian group. We shall show how to construct a field K and a valuation v of K with Γ as value group. Let k be any field and let $A = k[\Gamma]$ be the group algebra of Γ over k. By definition, A is freely generated as a k-vector space by elements x_α ($\alpha \in \Gamma$) such that $x_\alpha x_\beta = x_{\alpha+\beta}$. Show that A is an integral domain.
>
> If $u = \lambda_1 x_{\alpha_1} + \cdots + \lambda_n x_{\alpha_n}$ is any non-zero element of A, where the λ_i are all $\neq 0$ and $\alpha_1 < \cdots < \alpha_n$, define $v_0(u)$ to be α_1. Show that the mapping $v_0 : A - \{0\} \to \Gamma$ satisfies conditions (1) and (2) of Exercise 31.
>
> Let K be the field of fractions of A. Show that v_0 can be uniquely extended to a valuation v of K, and that the value group of v is precisely Γ.

Solution. Let $u = \sum_{i=1}^n \lambda_i x_{\alpha_i}$, $w = \sum_{j=1}^m \mu_j x_{\beta_j}$ be non-zero, with non-zero coefficients and $\alpha_1 < \cdots < \alpha_n$, $\beta_1 < \cdots < \beta_m$.

Domain and (1). In $uw = \sum \lambda_i\mu_j x_{\alpha_i+\beta_j}$, a pair $(i,j) \neq (1,1)$ has $\alpha_i + \beta_j > \alpha_1 + \beta_1$ (strict, since $\alpha_i \geq \alpha_1$, $\beta_j \geq \beta_1$ with one strict, and the order is compatible with addition). So the coefficient of $x_{\alpha_1+\beta_1}$ in uw is exactly $\lambda_1\mu_1 \neq 0$, all other terms having larger exponent. Hence $uw \neq 0$ and $v_0(uw) = \alpha_1 + \beta_1 = v_0(u) + v_0(w)$. So A is an integral domain and v_0 satisfies (1).

(2). If $u+w \neq 0$, its support is contained in $\{\alpha_i\}\cup\{\beta_j\}$, so its smallest exponent is $\geq \min(\alpha_1, \beta_1)$: $v_0(u+w) \geq \min(v_0(u), v_0(w))$.

Extension. Define $v(a/c) = v_0(a) - v_0(c)$ for $a, c \in A - \{0\}$. It is well defined: if $a/c = a'/c'$ then $ac' = a'c$, so $v_0(a) + v_0(c') = v_0(a') + v_0(c)$ by (1). Property (1) for v is clear. For (2): $a/c + b/d = (ad+bc)/cd$, so if the sum is $\neq 0$,

$$v\Big(\tfrac{a}{c}+\tfrac{b}{d}\Big) = v_0(ad+bc)-v_0(c)-v_0(d) \geq \min\big(v_0(a)+v_0(d),\, v_0(b)+v_0(c)\big)-v_0(c)-v_0(d) = \min\big(v(\tfrac{a}{c}), v(\tfrac{b}{d})\big).$$

Uniqueness. Any valuation v' of K extending v_0 satisfies $v'(a/c) = v'(a) - v'(c) = v_0(a) - v_0(c)$ by (1), so $v' = v$.

Value group. $v(x_\alpha) = \alpha$, so $v(K^*) \supseteq \Gamma$; and every $v(a/c) = v_0(a) - v_0(c) \in \Gamma$. Hence $v(K^*) = \Gamma$. □

Exercise 34

> **Problem.** Let A be a valuation ring and K its field of fractions. Let $f : A \to B$ be a ring homomorphism such that $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is a *closed* mapping. Then if $g : B \to K$ is any A-algebra homomorphism (i.e., if $g \circ f$ is the embedding of A in K) we have $g(B) = A$. [Let $C = g(B)$; obviously $C \supseteq A$. Let $\mathfrak{n}$ be a maximal ideal of C. Since f^* is closed, $\mathfrak{m} = \mathfrak{n} \cap A$ is the maximal ideal of A, whence $A_{\mathfrak{m}} = A$. Also the local ring $C_{\mathfrak{n}}$ dominates $A_{\mathfrak{m}}$. Hence by Exercise 27 we have $C_{\mathfrak{n}} = A$ and therefore $C \subseteq A$.]

Solution. Let $C = g(B)$; since gf is the inclusion, $A \subseteq C \subseteq K$. By Exercise 29, C is a local ring (indeed a local ring of A); let $\mathfrak{n}$ be its unique maximal ideal. Then $\mathfrak{N} = g^{-1}(\mathfrak{n})$ is a prime of B, in fact maximal ($B/\mathfrak{N} \cong C/\mathfrak{n}$), so $\{\mathfrak{N}\}$ is closed in $\operatorname{Spec} B$. As f^* is closed, $f^*(\{\mathfrak{N}\}) = \{f^{-1}(\mathfrak{N})\} = \{(gf)^{-1}(\mathfrak{n})\} = \{\mathfrak{n} \cap A\}$ is closed in $\operatorname{Spec} A$, i.e. $\mathfrak{m} = \mathfrak{n} \cap A$ is a maximal ideal of A (Ch. 1, Ex. 18(i)). Since A is local, $\mathfrak{m}$ is its maximal ideal and $A_{\mathfrak{m}} = A$. Now $C = C_{\mathfrak{n}}$ (local) contains A and $\mathfrak{n} \cap A = \mathfrak{m}$, i.e. C dominates A. By Exercise 27, a valuation ring is maximal for domination, so $C = A$. Hence $g(B) = A$. □

Exercise 35

> **Problem.** From Exercises 1 and 3 it follows that, if $f : A \to B$ is integral and C is any A-algebra, then the mapping $(f \otimes 1)^* : \operatorname{Spec}(B \otimes_A C) \to \operatorname{Spec}(C)$ is a closed map.
>
> Conversely, suppose that $f : A \to B$ has this property and that B is an integral domain. Then f is integral. [Replacing A by its image in B, reduce to the case where $A \subseteq B$ and f is the injection. Let K be the field of fractions of B and let A' be a valuation ring of K containing A. By (5.22) it is enough to show that A' contains B. By hypothesis $\operatorname{Spec}(B \otimes_A A') \to \operatorname{Spec}(A')$ is a closed map. Apply the result of Exercise 34 to the homomorphism $B \otimes_A A' \to K$ defined by $b \otimes a' \mapsto ba'$. It follows that $ba' \in A'$ for all $b \in B$ and all $a' \in A'$; taking $a' = 1$, we have what we want.]
>
> Show that the result just proved remains valid if B is a ring with only finitely many minimal prime ideals (e.g., if B is Noetherian). [Let $\mathfrak{p}_i$ be the minimal prime ideals. Then each composite homomorphism $A \to B \to B/\mathfrak{p}_i$ is integral, hence $A \to \prod(B/\mathfrak{p}_i)$ is integral, hence $A \to B/\mathfrak{N}$ is integral (where $\mathfrak{N}$ is the nilradical of B), hence finally $A \to B$ is integral.]

Solution. The direct statement. The ring homomorphism $C \to B \otimes_A C$, $c \mapsto 1 \otimes c$, is the base change of f. Apply Exercise 3 to the integral homomorphism $f : A \to B$ of A-algebras and the A-algebra C (with A itself in the role of the first algebra): $f \otimes 1 : A \otimes_A C = C \to B \otimes_A C$ is integral. By Exercise 1, $(f \otimes 1)^* : \operatorname{Spec}(B \otimes_A C) \to \operatorname{Spec}(C)$ is closed.

The converse for B a domain. Suppose $f : A \to B$ has the property: for every A-algebra C, $\operatorname{Spec}(B \otimes_A C) \to \operatorname{Spec} C$ is closed. Let $A_1 = f(A) \cong A/\operatorname{Ker} f$. The property is inherited by $A_1 \subseteq B$: an A_1-algebra C is an A-algebra and $B \otimes_{A_1} C = B \otimes_A C$ (since $A \to A_1$ is onto). And f integral means B integral over A_1. So we may assume $A \subseteq B$.

Let $K = \operatorname{Frac}(B)$ and let A' be any valuation ring of K containing A. Regard A' as an A-algebra. By hypothesis the map $\phi^* : \operatorname{Spec}(B \otimes_A A') \to \operatorname{Spec}(A')$ associated with $\phi : A' \to B \otimes_A A'$, $a' \mapsto 1 \otimes a'$, is closed. Define $g : B \otimes_A A' \to K$ by $b \otimes a' \mapsto ba'$ (a well-defined ring homomorphism: the map $(b, a') \mapsto ba'$ is A-bilinear). Then $g \circ \phi(a') = g(1 \otimes a') = a'$, the embedding of A' in K. By Exercise 34 (applied to the valuation ring A', the homomorphism ϕ and g), $g(B \otimes_A A') = A'$. Since $b = g(b \otimes 1)$ for $b \in B$, $B \subseteq A'$. As A' was an arbitrary valuation ring of K containing A, $B \subseteq \bigcap A' =$ the integral closure of A in K (5.22). So every element of B is integral over A, i.e. f is integral.

B with finitely many minimal primes. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the minimal primes of B, so $\mathfrak{N}(B) = \bigcap \mathfrak{p}_i$ (every prime contains a minimal prime, Ch. 1, Ex. 8). For each i the composite $f_i : A \to B/\mathfrak{p}_i$ again has the property: $(B/\mathfrak{p}_i) \otimes_A C = (B \otimes_A C)/\mathfrak{p}_i^e$, so $\operatorname{Spec}((B/\mathfrak{p}_i) \otimes_A C)$ is the closed subset $V(\mathfrak{p}_i^e)$ of $\operatorname{Spec}(B \otimes_A C)$ (Ch. 1, Ex. 21(iv)), and the restriction of a closed map to a closed subset is closed. Since $B/\mathfrak{p}_i$ is a domain, f_i is integral by the case just proved. Therefore

$A \to \prod_i B/\mathfrak{p}_i$ is integral (Exercise 6). The map $B/\mathfrak{N} \to \prod B/\mathfrak{p}_i$ is injective, and the image of A in $\prod B/\mathfrak{p}_i$ lies in $B/\mathfrak{N}$ (it is the image of A under $A \to B \to B/\mathfrak{N} \hookrightarrow \prod$); so each element of $B/\mathfrak{N}$ is integral, in $\prod B/\mathfrak{p}_i$, over the image of A, with a monic equation having coefficients in that image, which lies in $B/\mathfrak{N}$. Hence $A \to B/\mathfrak{N}$ is integral. Finally let $x \in B$; its class satisfies $\bar{x}^n + \bar{f}(a_1)\bar{x}^{n-1} + \cdots + \bar{f}(a_n) = 0$, i.e. $P(x) = x^n + f(a_1)x^{n-1} + \cdots + f(a_n) \in \mathfrak{N}(B)$, so $P(x)^N = 0$ for some N; P^N is monic with coefficients in $f(A)$, so x is integral over $f(A)$. Thus f is integral. $\square$